

PSP Data migration

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Overview

PSP is using Oracle RDS as datastore. PSP has two RDS instances running on AWS.

- Monolith - Contains all DT payroll realtime transactions data. Database size ~24 TB with growth rate ~3 TB/year
- IBOB - Contains mostly Audit data. Database size 4.5 TB with growth rate 0.5 TB/year

Above database would be migrated to Aurora PostgreSQL as below

- Full Load - Data is extracted from the source in a bulk extract manner and loaded directly into the target.
- CDC - The change capture process to replicate ongoing changes from a source to the target.

For rollback purpose, there would be other way replication.

Goal

- One time migration followed by replicating ongoing changes from Oracle Aurora PostgreSQL
- Enable reverse replication for rollback

Phases

Infra setup (IPS)	Schema Setup	Data Migration	Data Validation
• Deploy database infrastructure • Scale database infrastructure • Monitoring & alerting	• Initial DDL Schema conversion (Manual) • Use Schema generator	• Initial + Ongoing data migration (data replication) • Postgres Oracle replication	• Data Validation - One Time • Data Replication - CDC

Design Principles

- Data integrity - Zero data loss and no data corruption
- Near zero downtime approach for migration
- Select replication technology as per Intuit Technology selection principles
- Partition tables exceeding 300GB

Infra setup (IPS)

The design approach is "Infrastructure as Code". That is, all changes to the data store are initiated by code changes in github, not via the AWS API or the AWS Console.

IPS Automation provides:

- Create a new datastore
- Manage an existing datastore
- Add monitoring & alerting

We are planning to start with team managed approach where we will be using IPS automation to build database infra including monitoring and alerting. There are few issues/features pending currently in IPS. Once all are fixed and have confidence, we will switch fully to IPS.

[IPS - PSP Aurora PostgreSQL](#)

Schema Setup

Oracle RDS and Postgres RDS both are relational databases. However, there are differences in the data types and object definitions. We need to migrate Oracle schema objects to equivalent Postgres schema objects.

We used **AWS Schema Conversion Tool (SCT)** for initial/temporary schema conversion. It converts the source database schema and a majority of the database code objects, including views, stored procedures, and functions, to a format compatible with the target database. Any objects that cannot be automatically converted are clearly marked so that they can be manually converted to complete the migration.

- Data type mapping
- Tables
- Regular Indexes
- Function Based Indexes
- Primary Keys
- Foreign Keys
- Triggers
- Packages, Procedures & Functions
- Partitioning

Permanent solution would be to use schema generator to create schema in Postgres.

Data Migration

- Initial data migration
- CDC replication

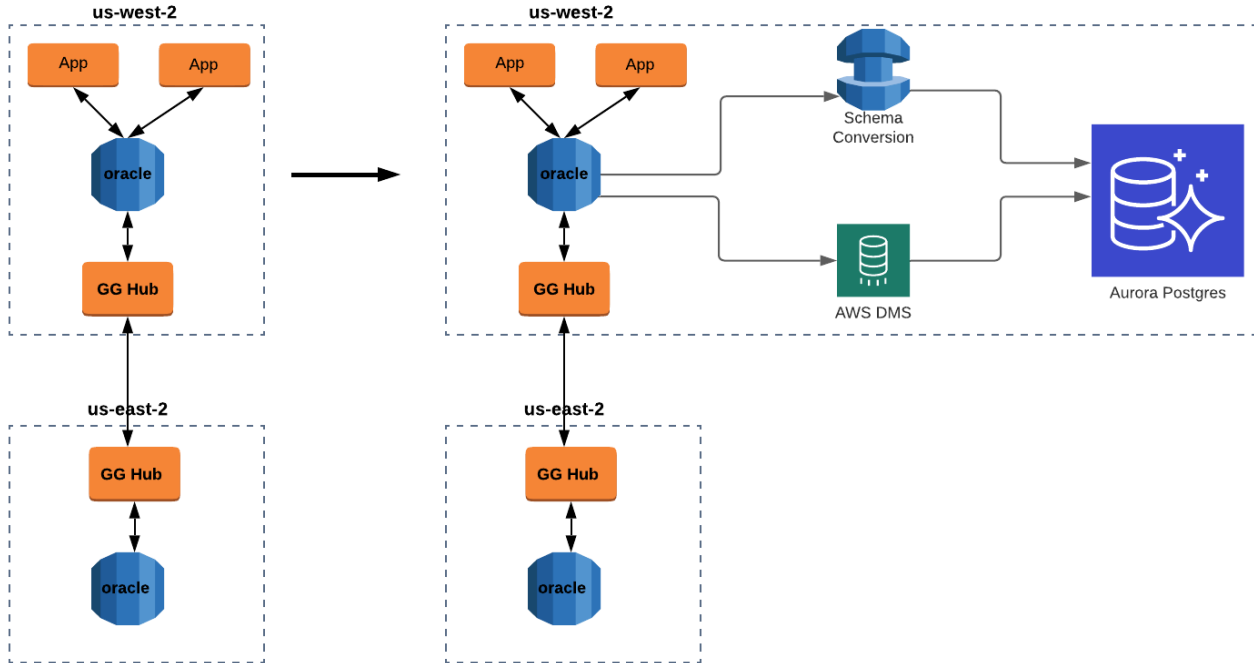
Once the schema is setup in the Postgres database, data can be copied to the target database. We are using **AWS Database Migration Service (AWS DMS)** for data replication. This supports heterogeneous migrations between different database platforms, such as Oracle or Microsoft SQL Server to Amazon Aurora.

This would be less downtime approach as source database remains fully operational during the migration. Downtime would be there only during cutover.

Data Replication

DMS

As per [Intuit Technology selection principles](#), we are going ahead with AWS native technology i.e. **AWS Database Migration Service (DMS)**.



Challenges

Currently PSP uses 8 range partitioned tables in oracle. When it comes to PostgreSQL partitioning functionality, there are certain limitations. We are hitting below major limitations of partitioning feature in PostgreSQL

1. Partitioning key has to be part of primary key on partitioned table. We do not see any change in this behaviour in future versions as well.
2. On partitioned table, Scope of Primary key constraint is restricted to partitions not global. Hence because of its scope is local, when we search based on this index, it scans all partitions of table.
3. Cannot add NOT VALID (NOT VALIDATED) foreign key constraint on partitioned table referring another partitioned/non-partitioned table.

Scope

Partitioned tables in scope:

	Table Name	Table Size (GB)	Primary Key	Partitioning key	Partition range	Remarks	Global Index Dependency
1	PSP_ENTRY_DETAIL_RECORD	915	ENTRY_DETAIL_RECORD_SEQ, REALM_ID	INITIATION_DATE	Monthly	Change partitioning to hash partition and start with ? #	
2	PSP_FINANCIAL_TRANSACTION	562	FINANCIAL_TRANSACTION_SEQ, REALM_ID	SETTLEMENT_DATE	Bi-monthly	Change partitioning to hash partition and start with ? #	• Data integrity via PK • FK dependencies • Query Performance via GI vs LI and partition pruning overhead
3	PSP_FINANCIAL_TRANSACTION_STATE	923	FINANCIAL_TRANS_STATE_SEQ, REALM_ID	TRANSACTION_STATE_EFF_DATE	Monthly	Change partitioning to hash partition and start with ? #	
4	PSP_LEDGER_BALANCE	157	LEDGER_BALANCE_SEQ, REALM_ID	BALANCE_DATE	Six-monthly	Change partitioning to hash partition and start with ? #	
5	PSP_MONEY_MOVEMENT_TRANSACTION	289	MONEY_MOVEMENT_TRANSACTION_SEQ, REALM_ID	INITIATION_DATE	Bi-monthly	Change partitioning to hash partition and start with ? #	
6	PSP_PAYCHECK	456	PAYCHECK_SEQ, REALM_ID	CREATED_DATE	Six-monthly	Change partitioning to hash partition and start with ? #	
7	PSP_PAYCHECK_SPLIT	285	PAYCHECK_SPLIT_SEQ, REALM_ID	CREATED_DATE	Six-monthly	Change partitioning to hash partition and start with ? #	

8	PSP_SOURCE_SYSTE M_TRANSMISSION		SOURCE_SYSTEM_TRANSM ISSION_SEQ, REALM_ID	CREATED_DATE	Monthly	No change in partitioning in APG in interest of scaling DMS replication	
---	------------------------------------	--	--	--------------	---------	--	--

Tables usage pattern

	Table Name	Filter pattern	Usage Scope (for eg: OLTP / Async jobs / Batch jobs etc)	Remarks
1	PSP_ENTRY_DETAIL_RECORD			
2	PSP_FINANCIAL_TRANSACTION			
3	PSP_FINANCIAL_TRANS_STATE			
4	PSP_LEDGER_BALANCE			
5	PSP_MONEY_MOVEMENT_TRANSACTION			
6	PSP_PAYCHECK			
7	PSP_PAYCHECK_SPLIT			
8	PSP_SOURCE_SYSTEM_TRANSMISSION			

Data Integrity

Environment : SYS

Table Name: PSP_FINANCIAL_TRANSACTION

DDL used : SCT

- On partitioned table, Scope of Primary key constraint is restricted to partitions. We were able to insert rows into 2 partitions with same primary key. Below is log:

DDL

```
CREATE TABLE pspadm.psp_financial_transaction(
  financial_transaction_seq CHARACTER VARYING(255) NOT NULL,
  version BIGINT NOT NULL,
  creator_id CHARACTER VARYING(30),
  created_date TIMESTAMP(6) WITHOUT TIME ZONE NOT NULL,
  modifier_id CHARACTER VARYING(30),
  modified_date TIMESTAMP(6) WITHOUT TIME ZONE NOT NULL,
  realm_id BIGINT NOT NULL DEFAULT - 1,
  financial_transaction_amount NUMERIC(19,4),
  settlement_date TIMESTAMP(6) WITHOUT TIME ZONE,
  settlement_type_cd CHARACTER VARYING(255),
  credit_bank_account_type CHARACTER VARYING(255),
  debit_bank_account_type CHARACTER VARYING(255),
  on_hold SMALLINT,
  sku CHARACTER VARYING(40),
  sku_quantity INTEGER,
  billing_detail_fk CHARACTER VARYING(255),
  credit_bank_account_fk CHARACTER VARYING(255),
  debit_bank_account_fk CHARACTER VARYING(255),
  company_fk CHARACTER VARYING(255),
  payroll_run_fk CHARACTER VARYING(255),
  paycheck_split_fk CHARACTER VARYING(255),
  transaction_type_fk CHARACTER VARYING(255) NOT NULL,
  law_fk CHARACTER VARYING(255),
  current_transaction_state_fk CHARACTER VARYING(255),
  original_transaction_fk CHARACTER VARYING(255),
  money_movement_transaction_fk CHARACTER VARYING(255),
  original_settlement_date TIMESTAMP(6) WITHOUT TIME ZONE,
  refund_type CHARACTER VARYING(255),
  comp_adjust_submission_fk CHARACTER VARYING(255),
```

```

tax_penalty_interest_fk CHARACTER VARYING(255),
relatable_transaction_fk CHARACTER VARYING(255),
bill_payment_split_fk CHARACTER VARYING(255),
company_law_fk CHARACTER VARYING(255),
status CHARACTER VARYING(255)
)
PARTITION BY RANGE (settlement_date)
WITH (
    OIDS=FALSE
);

```

Total partitions: 19

Partitions DDL:

```

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_9999
    PARTITION OF pspadm.psp_financial_transaction
    FOR VALUES FROM ('2022-01-01 00:00:00') TO (MAXVALUE);

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg12009
    PARTITION OF pspadm.psp_financial_transaction
    FOR VALUES FROM ('2009-01-01 00:00:00') TO ('2009-03-01 00:00:00');

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg12010
    PARTITION OF pspadm.psp_financial_transaction
    FOR VALUES FROM ('2010-01-01 00:00:00') TO ('2010-03-01 00:00:00');

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg22009
    PARTITION OF pspadm.psp_financial_transaction
    FOR VALUES FROM ('2009-03-01 00:00:00') TO ('2009-05-01 00:00:00');

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg22010
    PARTITION OF pspadm.psp_financial_transaction
    FOR VALUES FROM ('2010-03-01 00:00:00') TO ('2010-05-01 00:00:00');

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg22021
    PARTITION OF pspadm.psp_financial_transaction
    FOR VALUES FROM ('2011-01-01 00:00:00') TO ('2021-05-01 00:00:00');

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg32009
    PARTITION OF pspadm.psp_financial_transaction
    FOR VALUES FROM ('2009-05-01 00:00:00') TO ('2009-07-01 00:00:00');

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg32010
    PARTITION OF pspadm.psp_financial_transaction
    FOR VALUES FROM ('2010-05-01 00:00:00') TO ('2010-07-01 00:00:00');

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg32021
    PARTITION OF pspadm.psp_financial_transaction
    FOR VALUES FROM ('2021-05-01 00:00:00') TO ('2021-07-01 00:00:00');

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg42009
    PARTITION OF pspadm.psp_financial_transaction
    FOR VALUES FROM ('2009-07-01 00:00:00') TO ('2009-09-01 00:00:00');

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg42010
    PARTITION OF pspadm.psp_financial_transaction
    FOR VALUES FROM ('2010-07-01 00:00:00') TO ('2010-09-01 00:00:00');

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg42021
    PARTITION OF pspadm.psp_financial_transaction
    FOR VALUES FROM ('2021-07-01 00:00:00') TO ('2021-09-01 00:00:00');

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg52009
    PARTITION OF pspadm.psp_financial_transaction
    FOR VALUES FROM ('2009-09-01 00:00:00') TO ('2009-11-01 00:00:00');

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg52010
    PARTITION OF pspadm.psp_financial_transaction
    FOR VALUES FROM ('2010-09-01 00:00:00') TO ('2010-11-01 00:00:00');

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg52021

```

```

PARTITION OF pspadm.psp_financial_transaction
FOR VALUES FROM ('2021-09-01 00:00:00') TO ('2021-11-01 00:00:00');

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg62008
PARTITION OF pspadm.psp_financial_transaction
FOR VALUES FROM (MINVALUE) TO ('2009-01-01 00:00:00');

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg62009
PARTITION OF pspadm.psp_financial_transaction
FOR VALUES FROM ('2009-11-01 00:00:00') TO ('2010-01-01 00:00:00');

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg62010
PARTITION OF pspadm.psp_financial_transaction
FOR VALUES FROM ('2010-11-01 00:00:00') TO ('2011-01-01 00:00:00');

CREATE TABLE pspadm.psp_financial_transaction_financial_txn_mg62021
PARTITION OF pspadm.psp_financial_transaction
FOR VALUES FROM ('2021-11-01 00:00:00') TO ('2022-01-01 00:00:00');

```

create primary key

```

pspsys01=> ALTER TABLE pspadm.psp_financial_transaction_financial_txn_mg32010
pspsys01-> ADD CONSTRAINT sys_c0078697_financial_txn_mg32010 PRIMARY KEY (financial_transaction_seq, realm_id);
ALTER TABLE

pspsys01=> ALTER TABLE pspadm.psp_financial_transaction_financial_txn_mg22021
pspsys01-> ADD CONSTRAINT sys_c0078697_financial_txn_mg22021 PRIMARY KEY (financial_transaction_seq, realm_id);
ALTER TABLE

```

insert rows with same primary key

[illegible]

```

pspsys01=> alter table pspadm.psp_financial_transaction add constraint PSP_FINANCIAL_TRANSACTION_FK3 FOREIGN
KEY (debit_bank_account_fk, realm_id) references psp_bank_account (bank_account_seq, realm_id);
ERROR:  insert or update on table "psp_financial_transaction_financial_txn_mg22021" violates foreign key
constraint "psp_financial_transaction_fk3"
DETAIL:  Key (debit_bank_account_fk, realm_id)=(6fe7d904-da36-4afe-bd91-80d886272007, -1) is not present in
table "psp_bank_account".

pspsys01=>
pspsys01=> alter table pspadm.psp_financial_transaction add constraint PSP_FINANCIAL_TRANSACTION_FK3 FOREIGN KEY
("ORIGINAL_TRANSACTION_FK", "REALM_ID") REFERENCES pspadm.psp_financial_transaction
("FINANCIAL_TRANSACTION_SEQ", "REALM_ID");
ERROR:  cannot reference partitioned table "psp_financial_transaction"

```

Environment : SYS

Table Name: PSP_FINANCIAL_TRANSACTION

DDL used : SCT

When we create index on partitioned table and search based on that index, it scans all partitions of the table.

create index

```
pspsys01=> CREATE INDEX psp_financial_transaction_i5
pspsys01-> ON pspadm.psp_financial_transaction
pspsys01-> USING BTREE (company_fk ASC, current_transaction_state_fk ASC);
CREATE INDEX
pspsys01=>
```

Not visible at table level but could see when you describe partition of that table

```
pspsys01=> \d pspadm.psp_financial_transaction
?/? "pspadm.psp_financial_transaction"
      Column      |      Type
-----+-----
financial_transaction_seq | character varying(255)
version                | bigint
creator_id             | character varying(30)
created_date           | timestamp(6) without time zone
modifier_id            | character varying(30)
modified_date          | timestamp(6) without time zone
realm_id               | bigint
financial_transaction_amount | numeric(19,4)
settlement_date        | timestamp(6) without time zone
settlement_type_cd     | character varying(255)
credit_bank_account_type | character varying(255)
debit_bank_account_type | character varying(255)
on_hold                | smallint
sku                    | character varying(40)
sku_quantity           | integer
billing_detail_fk      | character varying(255)
credit_bank_account_fk | character varying(255)
debit_bank_account_fk  | character varying(255)
company_fk             | character varying(255)
payroll_run_fk         | character varying(255)
paycheck_split_fk      | character varying(255)
transaction_type_fk    | character varying(255)
law_fk                 | character varying(255)
current_transaction_state_fk | character varying(255)
original_transaction_fk | character varying(255)
money_movement_transaction_fk | character varying(255)
original_settlement_date | timestamp(6) without time zone
refund_type            | character varying(255)
comp_adjust_submission_fk | character varying(255)
tax_penalty_interest_fk | character varying(255)
relatable_transaction_fk | character varying(255)
bill_payment_split_fk  | character varying(255)
company_law_fk         | character varying(255)
status                 | character varying(255)

pspsys01=> \d pspadm.psp_financial_transaction_financial_txn_mg32010
Table "pspadm.psp_financial_transaction_financial_txn_mg32010"
      Column      |      Type      | Modifiers
-----+-----+-----
financial_transaction_seq | character varying(255) | not null
version                | bigint              | not null
creator_id             | character varying(30) |
created_date           | timestamp(6) without time zone | not null
modifier_id            | character varying(30) |
modified_date          | timestamp(6) without time zone | not null
```


realm_id	bigint	not null default '-1'::integer
financial_transaction_amount	numeric(19,4)	
settlement_date	timestamp(6) without time zone	
settlement_type_cd	character varying(255)	
credit_bank_account_type	character varying(255)	
debit_bank_account_type	character varying(255)	
on_hold	smallint	
sku	character varying(40)	
sku_quantity	integer	
billing_detail_fk	character varying(255)	
credit_bank_account_fk	character varying(255)	
debit_bank_account_fk	character varying(255)	
company_fk	character varying(255)	
payroll_run_fk	character varying(255)	
paycheck_split_fk	character varying(255)	
transaction_type_fk	character varying(255)	not null
law_fk	character varying(255)	
current_transaction_state_fk	character varying(255)	
original_transaction_fk	character varying(255)	
money_movement_transaction_fk	character varying(255)	
original_settlement_date	timestamp(6) without time zone	
refund_type	character varying(255)	
comp_adjust_submission_fk	character varying(255)	
tax_penalty_interest_fk	character varying(255)	
relatable_transaction_fk	character varying(255)	
bill_payment_split_fk	character varying(255)	
company_law_fk	character varying(255)	
status	character varying(255)	

Indexes:

"sys_c0078697_financial_txn_mg32010" PRIMARY KEY, btree (financial_transaction_seq, realm_id)

"psp_financial_transaction_fin_company_fk_current_transacti_idx9" btree (company_fk, current_transaction_state_fk)

Inherits: psp_financial_transaction

explain for select based on that index

```
pspsys01=> explain select count(*) from pspadm.psp_financial_transaction where company_fk='234' and
current_transaction_state_fk='123';
```

QUERY

PLAN

```
-----
Aggregate (cost=142.88..142.89 rows=1 width=8)
-> Append (cost=0.41..142.83 rows=19 width=0)
    -> Index Only Scan using psp_financial_transaction_fin_company_fk_current_transactio_idx on
    psp_financial_transaction_financial_txn_mg62008 (cost=0.41..8.42 rows=1 width=0)
        Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
    -> Index Only Scan using psp_financial_transaction_fin_company_fk_current_transacti_idx1 on
    psp_financial_transaction_financial_txn_mg12009 (cost=0.27..8.29 rows=1 width=0)
        Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
    -> Index Only Scan using psp_financial_transaction_fin_company_fk_current_transacti_idx2 on
    psp_financial_transaction_financial_txn_mg22009 (cost=0.14..7.36 rows=1 width=0)
        Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
    -> Seq Scan on psp_financial_transaction_financial_txn_mg32009 (cost=0.00..7.74 rows=1 width=0)
        Filter: (((company_fk)::text = '234'::text) AND ((current_transaction_state_fk)::text = '123'::
text))
    -> Seq Scan on psp_financial_transaction_financial_txn_mg42009 (cost=0.00..3.84 rows=1 width=0)
        Filter: (((company_fk)::text = '234'::text) AND ((current_transaction_state_fk)::text = '123'::
text))
    -> Seq Scan on psp_financial_transaction_financial_txn_mg52009 (cost=0.00..6.30 rows=1 width=0)
        Filter: (((company_fk)::text = '234'::text) AND ((current_transaction_state_fk)::text = '123'::
text))
    -> Seq Scan on psp_financial_transaction_financial_txn_mg62009 (cost=0.00..3.79 rows=1 width=0)
        Filter: (((company_fk)::text = '234'::text) AND ((current_transaction_state_fk)::text = '123'::
text))
    -> Index Only Scan using psp_financial_transaction_fin_company_fk_current_transacti_idx7 on
    psp_financial_transaction_financial_txn_mg12010 (cost=0.28..8.20 rows=1 width=0)
        Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
    -> Index Only Scan using psp_financial_transaction_fin_company_fk_current_transacti_idx8 on
    psp_financial_transaction_financial_txn_mg22010 (cost=0.28..8.29 rows=1 width=0)
        Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
    -> Index Only Scan using psp_financial_transaction_fin_company_fk_current_transacti_idx9 on
    psp_financial_transaction_financial_txn_mg32010 (cost=0.28..8.30 rows=1 width=0)
        Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
    -> Index Only Scan using psp_financial_transaction_fi_company_fk_current_transacti_idx10 on
    psp_financial_transaction_financial_txn_mg42010 (cost=0.28..8.29 rows=1 width=0)
        Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
    -> Seq Scan on psp_financial_transaction_financial_txn_mg52010 (cost=0.00..7.61 rows=1 width=0)
        Filter: (((company_fk)::text = '234'::text) AND ((current_transaction_state_fk)::text = '123'::
text))
    -> Index Only Scan using psp_financial_transaction_fi_company_fk_current_transacti_idx12 on
    psp_financial_transaction_financial_txn_mg62010 (cost=0.28..8.29 rows=1 width=0)
        Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
    -> Index Only Scan using psp_financial_transaction_fi_company_fk_current_transacti_idx13 on
    psp_financial_transaction_financial_txn_mg22021 (cost=0.56..8.58 rows=1 width=0)
        Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
    -> Index Only Scan using psp_financial_transaction_fi_company_fk_current_transacti_idx14 on
    psp_financial_transaction_financial_txn_mg32021 (cost=0.41..8.43 rows=1 width=0)
        Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
    -> Index Only Scan using psp_financial_transaction_fi_company_fk_current_transacti_idx15 on
    psp_financial_transaction_financial_txn_mg42021 (cost=0.41..8.43 rows=1 width=0)
        Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
    -> Seq Scan on psp_financial_transaction_financial_txn_mg52021 (cost=0.00..6.28 rows=1 width=0)
        Filter: (((company_fk)::text = '234'::text) AND ((current_transaction_state_fk)::text = '123'::
text))
    -> Seq Scan on psp_financial_transaction_financial_txn_mg62021 (cost=0.00..7.84 rows=1 width=0)
        Filter: (((company_fk)::text = '234'::text) AND ((current_transaction_state_fk)::text = '123'::
text))
    -> Index Only Scan using psp_financial_transaction_fi_company_fk_current_transacti_idx18 on
    psp_financial_transaction_financial_txn_9999 (cost=0.41..8.43 rows=1 width=0)
        Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
(40 rows)
```

```
pspsys01=> explain (analyze,buffers) select count(*) from pspadm.psp_financial_transaction where
company_fk='234' and current_transaction_state_fk='123';
```

QUERY

PLAN

```
-----
--
Aggregate (cost=142.88..142.89 rows=1 width=8) (actual time=0.291..0.291 rows=1 loops=1)
  Buffers: shared hit=63
  -> Append (cost=0.41..142.83 rows=19 width=0) (actual time=0.289..0.289 rows=0 loops=1)
    Buffers: shared hit=63
    -> Index Only Scan using psp_financial_transaction_fin_company_fk_current_transactio_idx on
        psp_financial_transaction_financial_txn_mg62008 (cost=0.41..8.42 rows=1 width=0) (actual time=0.014..0.014
        rows=0 loops=1)
      Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
      Heap Fetches: 0
      Buffers: shared hit=3
    -> Index Only Scan using psp_financial_transaction_fin_company_fk_current_transacti_idx1 on
        psp_financial_transaction_financial_txn_mg12009 (cost=0.27..8.29 rows=1 width=0) (actual time=0.008..0.008
        rows=0 loops=1)
      Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
      Heap Fetches: 0
      Buffers: shared hit=2
    -> Index Only Scan using psp_financial_transaction_fin_company_fk_current_transacti_idx2 on
        psp_financial_transaction_financial_txn_mg22009 (cost=0.14..7.36 rows=1 width=0) (actual time=0.006..0.006
        rows=0 loops=1)
      Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
      Heap Fetches: 0
      Buffers: shared hit=1
    -> Seq Scan on psp_financial_transaction_financial_txn_mg32009 (cost=0.00..7.74 rows=1 width=0)
        (actual time=0.038..0.038 rows=0 loops=1)
      Filter: (((company_fk)::text = '234'::text) AND ((current_transaction_state_fk)::text = '123'::
      text))
      Rows Removed by Filter: 116
      Buffers: shared hit=6
    -> Seq Scan on psp_financial_transaction_financial_txn_mg42009 (cost=0.00..3.84 rows=1 width=0)
        (actual time=0.015..0.015 rows=0 loops=1)
      Filter: (((company_fk)::text = '234'::text) AND ((current_transaction_state_fk)::text = '123'::
      text))
      Rows Removed by Filter: 56
      Buffers: shared hit=3
    -> Seq Scan on psp_financial_transaction_financial_txn_mg52009 (cost=0.00..6.30 rows=1 width=0)
        (actual time=0.026..0.026 rows=0 loops=1)
      Filter: (((company_fk)::text = '234'::text) AND ((current_transaction_state_fk)::text = '123'::
      text))
      Rows Removed by Filter: 87
      Buffers: shared hit=5
    -> Seq Scan on psp_financial_transaction_financial_txn_mg62009 (cost=0.00..3.79 rows=1 width=0)
        (actual time=0.015..0.015 rows=0 loops=1)
      Filter: (((company_fk)::text = '234'::text) AND ((current_transaction_state_fk)::text = '123'::
      text))
      Rows Removed by Filter: 53
      Buffers: shared hit=3
    -> Index Only Scan using psp_financial_transaction_fin_company_fk_current_transacti_idx7 on
        psp_financial_transaction_financial_txn_mg12010 (cost=0.28..8.20 rows=1 width=0) (actual time=0.007..0.007
        rows=0 loops=1)
      Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
      Heap Fetches: 0
      Buffers: shared hit=2
    -> Index Only Scan using psp_financial_transaction_fin_company_fk_current_transacti_idx8 on
        psp_financial_transaction_financial_txn_mg22010 (cost=0.28..8.29 rows=1 width=0) (actual time=0.007..0.007
        rows=0 loops=1)
      Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
      Heap Fetches: 0
      Buffers: shared hit=2
    -> Index Only Scan using psp_financial_transaction_fin_company_fk_current_transacti_idx9 on
        psp_financial_transaction_financial_txn_mg32010 (cost=0.28..8.30 rows=1 width=0) (actual time=0.008..0.008
        rows=0 loops=1)
```

```

Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
Heap Fetches: 0
Buffers: shared hit=2
-> Index Only Scan using psp_financial_transaction_fi_company_fk_current_transacti_idx10 on
psp_financial_transaction_financial_txn_mg42010 (cost=0.28..8.29 rows=1 width=0) (actual time=0.008..0.008
rows=0 loops=1)
Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
Heap Fetches: 0
Buffers: shared hit=2
-> Seq Scan on psp_financial_transaction_financial_txn_mg52010 (cost=0.00..7.61 rows=1 width=0)
(actual time=0.030..0.030 rows=0 loops=1)
Filter: (((company_fk)::text = '234'::text) AND ((current_transaction_state_fk)::text = '123'::
text))
Rows Removed by Filter: 107
Buffers: shared hit=6
-> Index Only Scan using psp_financial_transaction_fi_company_fk_current_transacti_idx12 on
psp_financial_transaction_financial_txn_mg62010 (cost=0.28..8.29 rows=1 width=0) (actual time=0.007..0.007
rows=0 loops=1)
Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
Heap Fetches: 0
Buffers: shared hit=2
-> Index Only Scan using psp_financial_transaction_fi_company_fk_current_transacti_idx13 on
psp_financial_transaction_financial_txn_mg22021 (cost=0.56..8.58 rows=1 width=0) (actual time=0.013..0.013
rows=0 loops=1)
Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
Heap Fetches: 0
Buffers: shared hit=4
-> Index Only Scan using psp_financial_transaction_fi_company_fk_current_transacti_idx14 on
psp_financial_transaction_financial_txn_mg32021 (cost=0.41..8.43 rows=1 width=0) (actual time=0.011..0.011
rows=0 loops=1)
Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
Heap Fetches: 0
Buffers: shared hit=3
-> Index Only Scan using psp_financial_transaction_fi_company_fk_current_transacti_idx15 on
psp_financial_transaction_financial_txn_mg42021 (cost=0.41..8.43 rows=1 width=0) (actual time=0.010..0.010
rows=0 loops=1)
Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
Heap Fetches: 0
Buffers: shared hit=3
-> Seq Scan on psp_financial_transaction_financial_txn_mg52021 (cost=0.00..6.28 rows=1 width=0)
(actual time=0.026..0.026 rows=0 loops=1)
Filter: (((company_fk)::text = '234'::text) AND ((current_transaction_state_fk)::text = '123'::
text))
Rows Removed by Filter: 85
Buffers: shared hit=5
-> Seq Scan on psp_financial_transaction_financial_txn_mg62021 (cost=0.00..7.84 rows=1 width=0)
(actual time=0.032..0.032 rows=0 loops=1)
Filter: (((company_fk)::text = '234'::text) AND ((current_transaction_state_fk)::text = '123'::
text))
Rows Removed by Filter: 123
Buffers: shared hit=6
-> Index Only Scan using psp_financial_transaction_fi_company_fk_current_transacti_idx18 on
psp_financial_transaction_financial_txn_9999 (cost=0.41..8.43 rows=1 width=0) (actual time=0.008..0.009 rows=0
loops=1)
Index Cond: ((company_fk = '234'::text) AND (current_transaction_state_fk = '123'::text))
Heap Fetches: 0
Buffers: shared hit=3
Planning Time: 0.718 ms
Execution Time: 0.381 ms
(82 rows)

```

Data Validation

Partitioned Tables

Since partitioned table does not have global primary key, DMS does not compare with message "No Primary Key".

Tables with CLOB column

How to make sure CLOB data is migrated and validated successfully ?

Table with LOB Columns

	TABLE_NAME	COLUMNS	Validation Status
1	AT_WIDGET	WIDGET_CONFIGURATION	
2	PSP_ACHENROLLMENT_FILE	FILE_CONTENT, FILE_CONTENT_ENC	OOS CLOB column: file_content, Failure type: record_diff
3	PSP_CHECK_PRINT_SIGNATURE	SIGNATURE	
4	PSP_ENTITLEMENT_MESSAGE	MESSAGE, MESSAGE_ENC	
5	PSP_ENTITY_UPDATE	CHANGED_ATTRIBUTES	
6	PSP_LEDGER_OPERATION_JOB	ORIGINAL_FILE, PROCESSED_FILE	OOS CLOB columns: (original_file, processed_file), Failure type: record_diff
7	PSP_MESSAGE_LOG	REQUEST_LOG, RESPONSE_LOG	
8	PSP_RTBAUTOMATIONBACKUP	RTB_BACKUP	
9	PSP_SAVED_REPORTS	QUERY	
10	PSP_SOURCESYS_PRINTEDCHK_IN FO	BANK_LOGO, SOURCE_SYSTEM_LOGO	
11	PSP_STATE_REPORT_OUTPUT	REPORT_OUTPUT	OOS CLOB column: report_output, Failure type: record_diff
12	PSP_SUICREDITS_JOB	PROCESSED_FILE	
13	PSP_TAX_CREDITS9061	FORM9061	
14	PSP_TAX_CREDITS_APPLICATION	SIGNED_DOCUMENT, UNSIGNED_DOCUMENT	

DMS Validation shows failures for below tables for CLOB columns

Data model optimizations for PostgreSQL

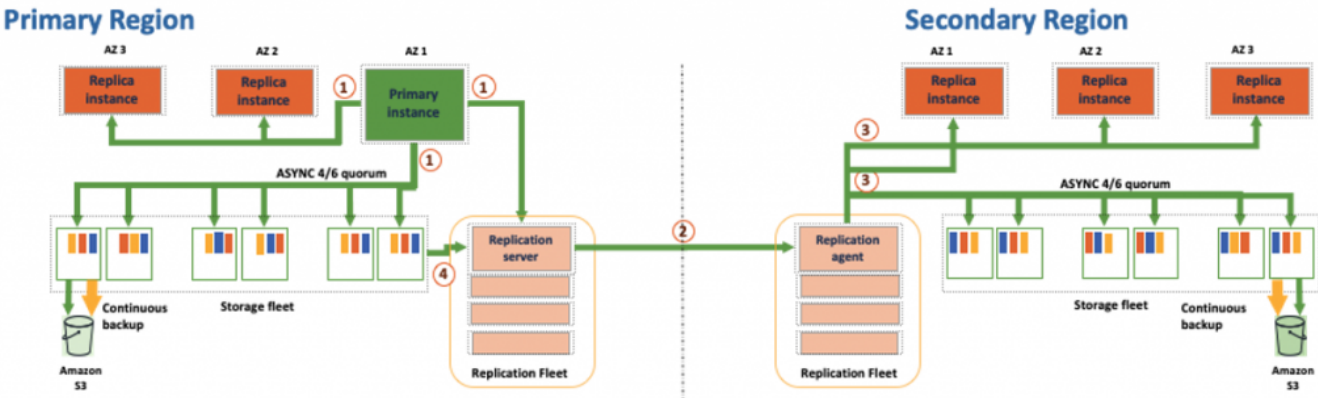
Design and POC - Yet to start

Query Tuning for PostgreSQL

Design and POC - Yet to start

Aurora PostgreSQL HA & DR

- Planning to use Aurora Global database for HA and DR



Data lake integration from Aurora PostgreSQL

Design and POC - Yet to start